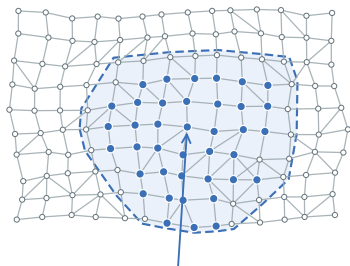
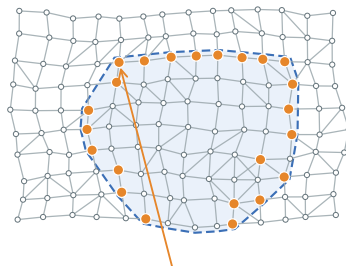


(a) Region interior



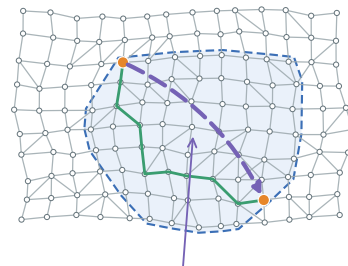
Interior nodes removed

(b) Boundary nodes



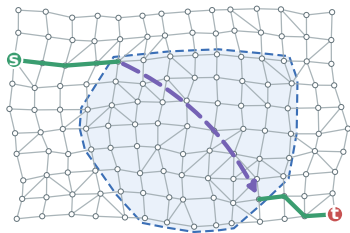
Boundary nodes retained

(c) Materialized shortcut

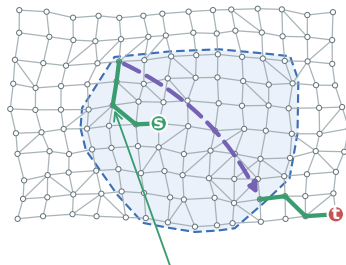


Overlay shortcut = exact path distance

(d) Both endpoints outside

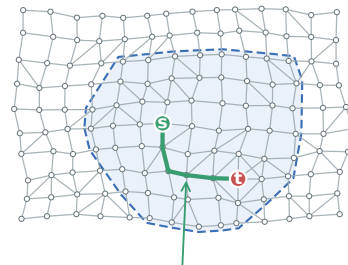


(e) One endpoint inside



Local endpoint access

(f) Both endpoints in one region



Exact in-region path

— Road edge

• Interior node

• Boundary node

— Materialized shortcut (overlay)

• Origin s

• Destination t